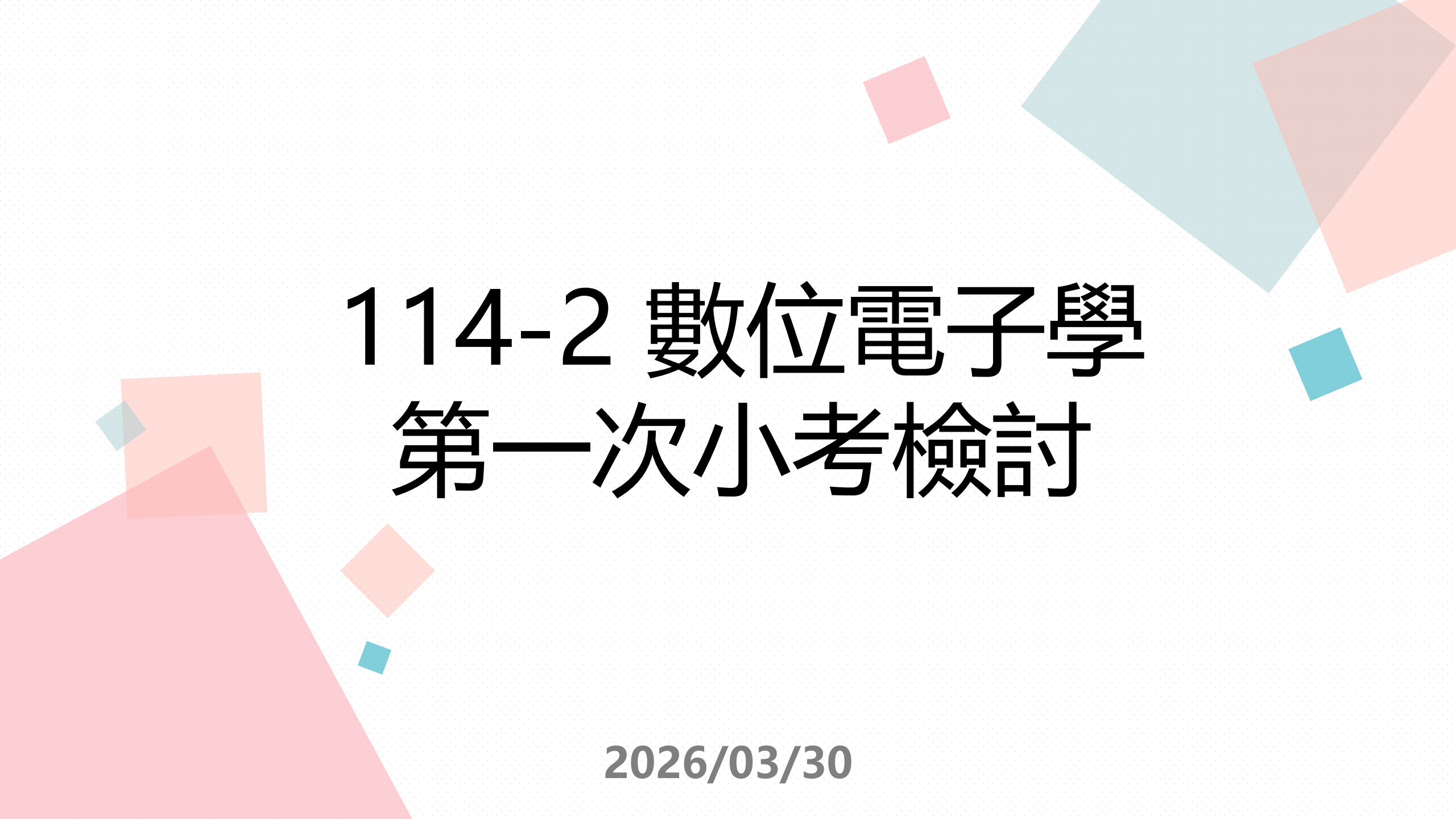


The background features several overlapping geometric shapes, primarily squares and triangles, in shades of pink, teal, and light blue. These shapes are scattered across the slide, with some appearing as large, semi-transparent overlays and others as smaller, solid-colored elements.

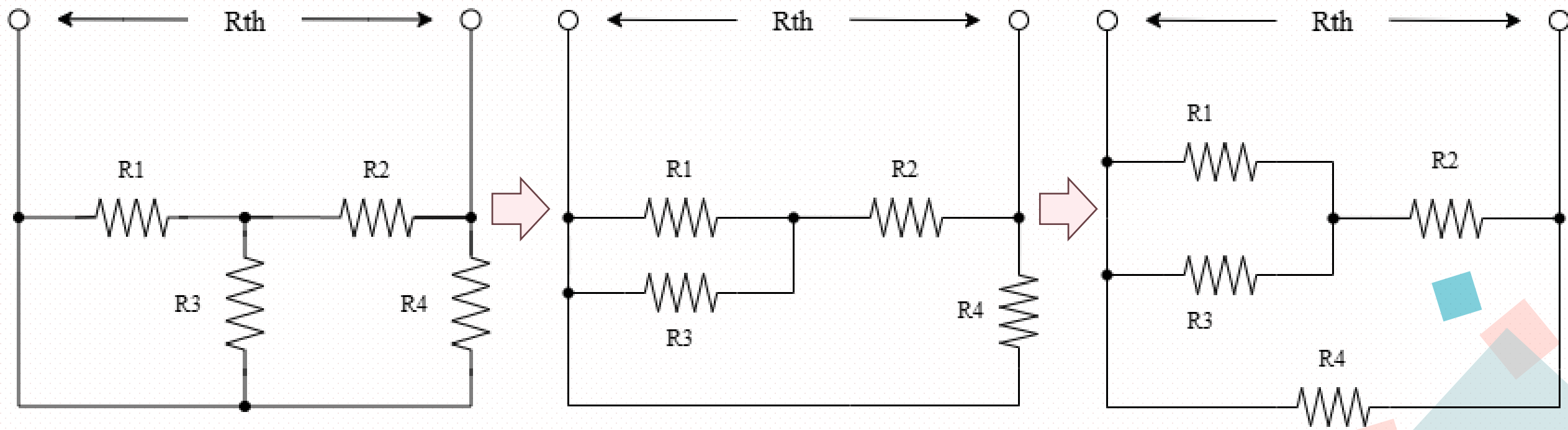
114-2 數位電子學

第一次小考檢討

2026/03/30

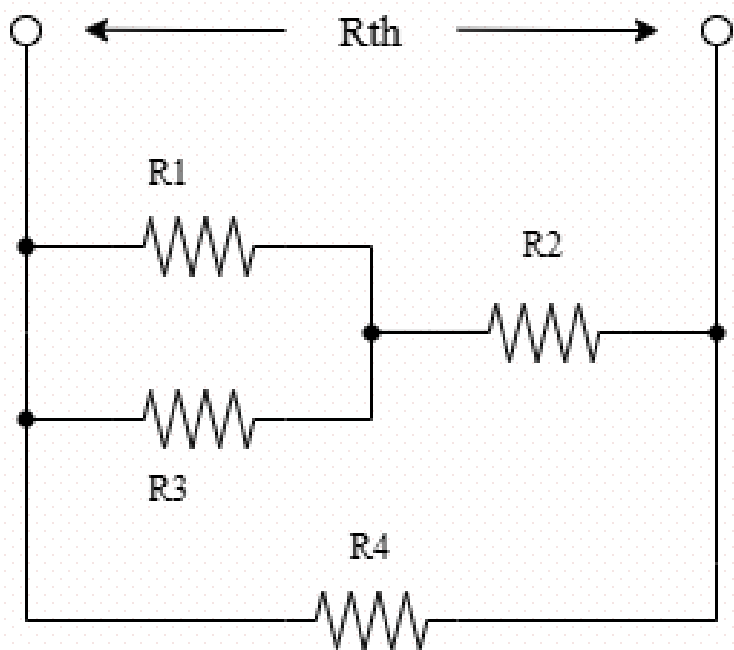
第一題

做電路化簡



第一題

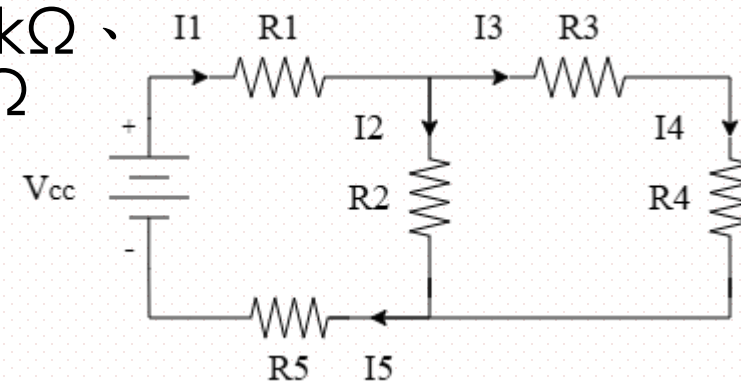
做電路化簡



$$\begin{aligned} R_{th} &= ((R1 // R3) + R2) // R4 \\ &= ((3 // 9) + 9) // 9 \\ &= ((\frac{9}{4}) + 9) // 9 \\ &= (\frac{45}{4}) // 9 \\ &= 5 \text{ (k}\Omega\text{)} \end{aligned}$$

第二題

$V_{CC} = 12V$ 、 $R1 = 3k\Omega$ 、 $R2 = 3k\Omega$ 、
 $R3 = 4k\Omega$ 、 $R4 = 2k\Omega$ 、 $R5 = 7k\Omega$



$$\begin{aligned} R_{\text{total}} &= R1 + (R2 // (R3 + R4)) + R5 \\ &= 3 + (3 // 6) + 7 \\ &= 3 + 2 + 7 = 12 \text{ (k}\Omega\text{)} \end{aligned}$$

$$I1 = I5 = V_{CC} / R_{\text{total}} = 12 / 12 = 1 \text{ (mA)}$$

$$V1 = I1 * R1 = 3 \text{ (V)}$$

$$V5 = I5 * R5 = 7 \text{ (V)}$$

$$V2 = 12 - 3 - 7 = 2 \text{ (V)}$$

$$I2 = V2 / R2 = 2/3 \text{ (mA)}$$

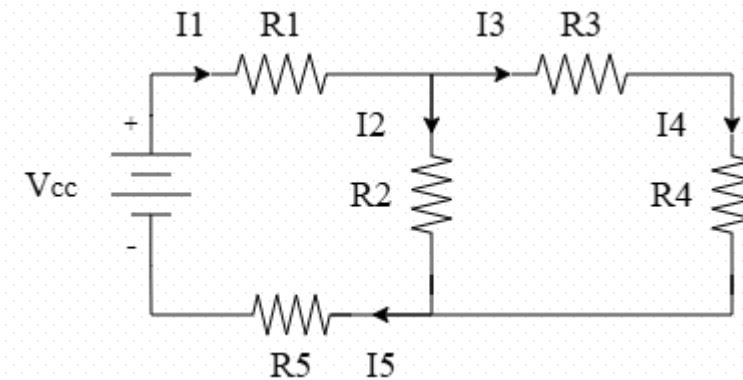
$$I3 = V2 / (R3 + R4) = 2/6 = 1/3 \text{ (mA)}$$

$$I4 = I3 = 1/3 \text{ (mA)}$$

$$V3 = I3 * R3 = 4/3 \text{ (V)}$$

$$V4 = I4 * R4 = 2/3 \text{ (V)}$$

第二題_Ans



I1	I2	I3	I4	I5
1	$\frac{2}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	1

單位：mA

V1	V2	V3	V4	V5
3	2	$\frac{4}{3}$	$\frac{2}{3}$	7

單位：V

$$I = 5\cos 2t \text{ (A)} , R = 4\Omega , C = 3F , L = 6H$$

第三題

- $V_R = I * R = 5\cos 2t * 4 = 20\cos 2t \text{ (V)}$
- $V_L = L * \frac{dI(t)}{dt} = 6 * \frac{d 5\cos 2t}{dt} = 6 * 5 * (-2\sin 2t) = -60\sin 2t \text{ (V)}$
- $V_C = \frac{1}{C} \int I(t) dt = \frac{1}{3} \int 5\cos 2t dt = \frac{1}{3} * 5 * \left(\frac{1}{2}\sin 2t\right) = \frac{5}{6}\sin 2t \text{ (V)}$

第三題

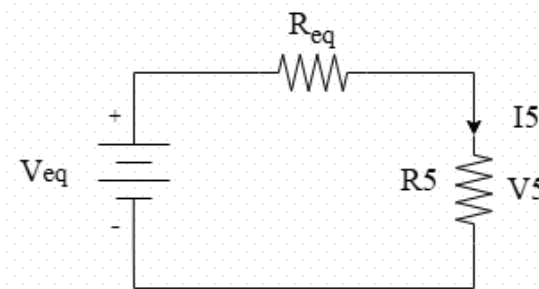
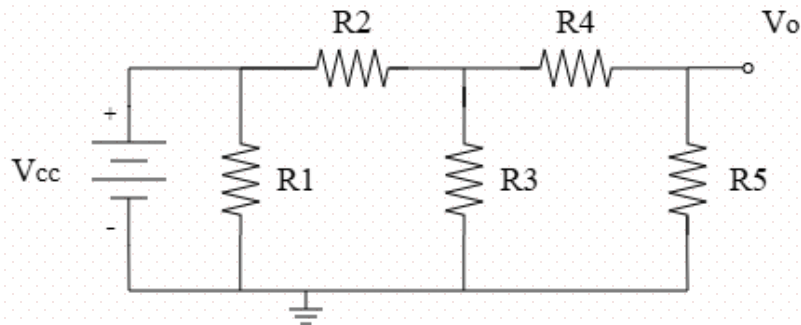
$$\frac{d}{dt} \sin(at) = a \cos(at)$$

$$\int \sin(at) dt = -\frac{1}{a} \cos(at)$$

$$\frac{d}{dt} \cos(at) = -a \sin(at)$$

$$\int \cos(at) dt = \frac{1}{a} \sin(at)$$

第四題

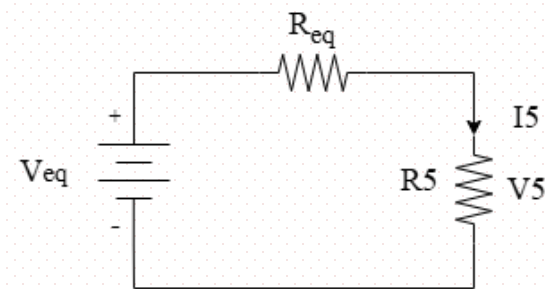
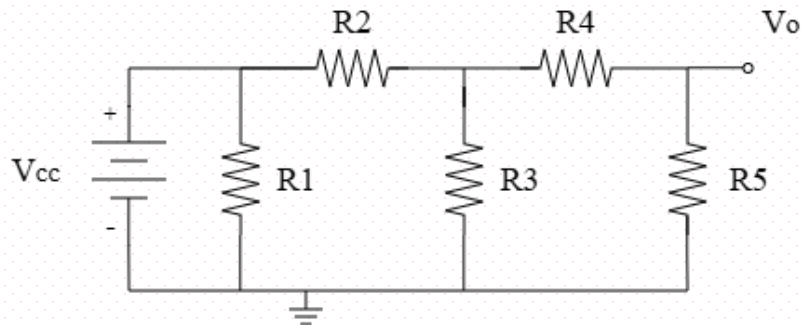


V_{eq} :

- 先將 R_5 移走，流經 R_4 的電流為零
- R_1 與 V_{cc} 並聯，不會影響右側的電壓分佈
- 直接對 R_2 與 R_3 用分壓計算

- $$V_{eq} = V_{cc} * \frac{R_3}{R_2 + R_3} = 12 * \frac{6}{6 + 6} = 6V$$

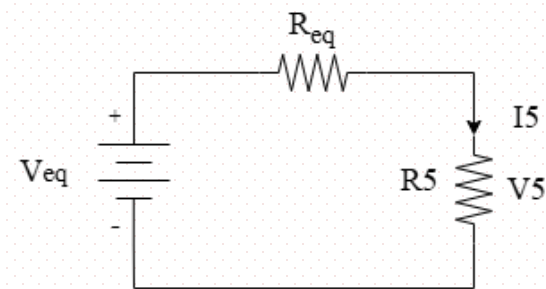
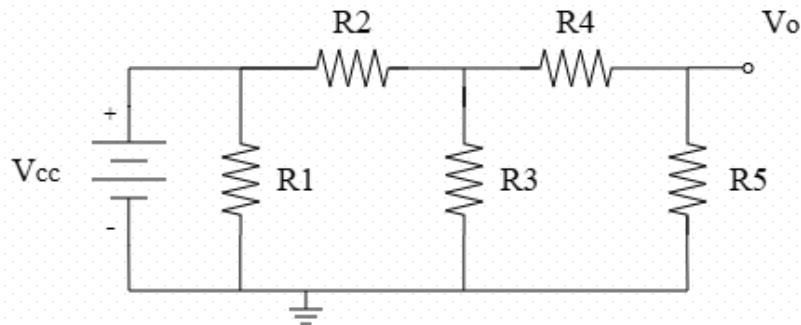
第四題



Req :

- 將 V_{cc} 短路， $R1$ 被短路忽略
- 從 V_o 向左進入，電流會先經過 $R4$ ，接著分別流經 $R2$ 與 $R3$
- $R_{eq} = R4 + (R2 // R3) = 5 + (6 // 6) = 5 + 3 = 8\Omega$

第四題



$$V_{eq} = 6V, R_{eq} = 8\Omega$$

- $I_5 = \frac{V_{eq}}{R_{eq} + R_5} = \frac{6}{8 + 4} = 0.5A$
- $V_5 = I_5 * R_5 = 0.5 * 4 = 2V$

第五題

$$V_{CC} = 100V, R = 12.5K\Omega, C = 80\mu F$$

(1) V_C 由 0V 充電至 80V

(2) V_C 由 100V 放電至 10V

(1) 充電：

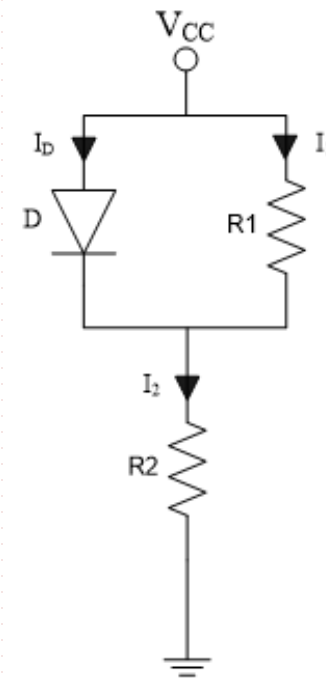
$$\begin{aligned} t &= -RC * \ln\left(1 - \frac{V_C}{V_{CC}}\right) \\ &= -12.5 * 80 * \ln(0.2) \\ &= 1.6 \text{ (s)} \end{aligned}$$

(2) 放電：

$$\begin{aligned} t &= -RC * \ln\left(\frac{V_C}{V_{CC}}\right) \\ &= -12.5 * 80 * \ln(0.1) \\ &= 2.3 \text{ (s)} \end{aligned}$$

第六題

$V_{CC} = 2.7V$ ， $R_2 = 20\Omega$ ，若(1) $R_1 = 14\Omega$
分別求出二極體的電壓 V_D 和電流 I_D



假設on：

$$V_D = 0.7V = V_1$$

$$I_1 = \frac{V_1}{R_1} = \frac{0.7}{14} = 0.05A$$

$$I_2 = \frac{V_{CC} - V_1}{R_2} = \frac{2.7 - 0.7}{20} = \frac{2}{20} = 0.1A$$

$$I_D = I_2 - I_1 = 0.1 - 0.05 = 0.05A$$

假設off：

$$I_D = 0A$$

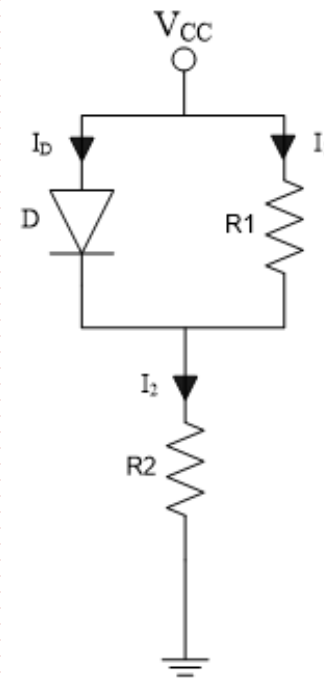
$$V_D = V_{CC} * \frac{R_1}{R_1 + R_2}$$

$$= 2.7 * \frac{14}{14 + 20}$$

$$\approx 1.1V > V_{cut-in} (0.5V)$$

第六題

$V_{CC} = 2.7V$ ， $R_2 = 20\Omega$ ，若(2) $R_1 = 2.5\Omega$
分別求出二極體的電壓 V_D 和電流 I_D



假設on：

$$V_D = 0.7V = V_1$$

$$I_1 = \frac{V_1}{R_1} = \frac{0.7}{2.5} = 0.28A$$

$$I_2 = \frac{V_{CC} - V_1}{R_2} = \frac{2.7 - 0.7}{20} = \frac{2}{20} = 0.1A$$

$$I_D = I_2 - I_1 = 0.1 - 0.28 < 0 \rightarrow \leftarrow$$

假設off：

$$I_D = 0A$$

$$\begin{aligned} V_D &= V_{CC} * \frac{R_1}{R_1 + R_2} \\ &= 2.7 * \frac{2.5}{2.5 + 20} \\ &= 2.7 * \frac{2.5}{22.5} \\ &= 2.7 * \frac{1}{9} \\ &= 0.3V \end{aligned}$$